

BDA Practical No.4

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Step 1:-Type the command given below:-

```
stop-all.sh
```

Step 2:-Now type:-

```
hdfs namenode -format
```

Step 3:-Now type:-

```
start-dfs.sh
```

Step 4:-Type:-

```
start-yarn.sh
```

Note:-If you face any errors in Step No.3 and 4 then refer BDA Practical No.1 pdf by Aditya Nikam

Step 5:-Type:-

```
jps
```

Step 6:-Now make directory named “ship” by typing the below command:-

```
mkdir ship
```

Step 7:-Go to directory named “ship”

```
cd ship
```

Step 8:-Now type the command given below:-

```
nano titanic.csv
```

And now paste the code given below in the titanic.csv file:-

```
PassengerId,Survived,Pclass,Name,Sex,Age
```

```
1,0,3,John,male,22
```

```
2,1,1,Mary,female,38
```

```
3,1,3,Anna,female,26
```

```
4,0,1,Bob,male,35
```

5,0,2,Lily,female,30

6,0,3,Susan,female,18

Now press Ctrl + O, then press Enter and then press Ctrl + X

Step 8:- Now type the command given below:-

```
nano mapper.py
```

And now paste the code given below in the mapper.py file:-

```
#!/usr/bin/env python3
```

```
import sys
```

```
for line in sys.stdin:
```

```
    if line.startswith("PassengerId"):
```

```
        continue
```

```
    fields = line.strip().split(',')
```

```
    if len(fields) < 6:
```

```
        continue
```

```
    survived = fields[1]
```

```
    pclass = fields[2]
```

```
    sex = fields[4].lower()
```

```
    age = fields[5]
```

```
    if survived == '0': # only for those who died
```

```
        if sex == 'male' and age:
```

```
            print(f"male\t{age}")
```

```
        elif sex == 'female':
```

```
            print(f"class_{pclass}\t1")
```

Now press Ctrl + O, then press Enter and then press Ctrl + X

Step 9:- Now type the command given below:-

```
nano reducer.py
```

And now paste the code given below in the reducer.py file:-

```
#!/usr/bin/env python3

import sys

male_age_sum = 0

male_count = 0

class_deaths = {}

for line in sys.stdin:

    key, value = line.strip().split('\t')

    if key == "male":

        try:

            age = float(value)

            male_age_sum += age

            male_count += 1

        except ValueError:

            continue

    elif key.startswith("class_"):

        if key in class_deaths:

            class_deaths[key] += 1

        else:

            class_deaths[key] = 1

if male_count > 0:

    avg_age = male_age_sum / male_count

    print(f"Average age of deceased males: {avg_age:.2f}")

for cls in sorted(class_deaths):

    formatted_class = cls.replace('class_', 'Class ')

    print(f"{formatted_class} female deaths: {class_deaths[cls]}")
```

Now press Ctrl + O, then press Enter and then press Ctrl + X

Step 10:- Now type the command given below:-

```
chmod +x mapper.py
```

Step 11:-Now type:-

```
chmod +x reducer.py
```

Step 12:-Now type:-

```
hadoop fs -mkdir -p /input_titanic
```

Step 13:-Now type:-

```
hadoop fs -put titanic.csv /input_titanic/
```

Step 14:-Type the command given below:-

```
find /usr /opt ~ -name "hadoop-streaming*.jar" 2>/dev/null
```

Output will be something like below:-

```
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-test-sources.jar
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-sources.jar
/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
```

So see here you will get the path and version of hadoop streaming like see in the above output /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar is hadoop streaming version and path. We will need this path in step 8 (Don't see/copy test_sources.jar or sources.jar file remember that). And after pasting the path ensure that you type forward slash at the end “\” in the first line of the above command as you can see.

Step 19:-Now type:-

```
hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-files mapper.py, reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
```

```
-input /input_titanic/titanic.csv \
```

```
-output /output_titanic
```

You will see a lot of "INFO" text rapidly scrolling on your screen.Wait until it finishes completely.

See here in this command given above the first line has will have the path and version number of hadoop streaming we got from Step No.17 so we that pasted that path after hadoop jar in the first line of the above command so your path may be little different depending on version number note this

Also ensure that you have named files as "mapper.py" and "reducer.py" if the names are different then replace the names in the above command otherwise above command will give error.

Step 19:-Now type the below command:-

```
hadoop fs -cat /output_titanic/part-00000
```

You will see the output like below:-

Average age of deceased males: 28.50

Class 2 female deaths: 1

Class 3 female deaths: 1

If the output is like above means your practical is completed

